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Decoding defensive performance: a machine learning approach to football player valuation

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Abstract

Analyzing defensive actions, which have traditionally received less attention than offensive metrics, is a significant challenge in football analytics. This research presents an innovative methodology that utilizes XGBoost and deep neural networks to evaluate defensive performance using metrics such as On-Ball Value (OBV), Valuing Actions by Estimating Probabilities (VAEP), and eXpected Threat (xT). The study proposes a machine learning-based framework for evaluating the value of defensive players. A case study using expert ratings and market values from the Polish PKO BP Ekstraklasa demonstrates the method's effectiveness. The results advance the field of sports analytics by addressing the persistent problem of accurately valuing the defensive contributions of football players.

Keywords Football, Player evaluation, Machine learning, Deep learning, XGBoost

Introduction

One of the world's most popular sports, football transcends mere physical prowess to embody intricate tactical maneuvers. It is essential to understand that football is not just a physical endeavor but also an intellectual one, where defensive strategies play a significant role [1].

Defensive tactics in football may lack the glamour often associated with offensive prowess, but they are just as crucial to a team's triumph [2]. A solid defensive structure thwarts the opposition's attacks and lays the groundwork for adequate counterplay. Teams that prioritize defensive solidity demonstrate skillful organization, discipline, and adaptability. Contemporary football, especially at the elite level, shows the evolution of defensive tactics. From conventional defensive formations, such as the 1–4–4–2, to modern tactical systems, including pressing or 'parking the bus,' the spectrum of defensive strategies is diverse and sophisticated [3]. One notable example is the renowned defensive prowess of Italian football teams, which has often translated into international success. At the heart of defensive tactics is keen anticipation and understanding of the opponent's intentions, which requires physical strength and tactical knowledge. Players must quickly grasp the game's dynamics, communicate effectively, and collaborate to devise effective defensive strategies. More than just

goalkeeping, defensive tactics dictate the tempo of the game, the management of the pitch, and the exploitation of opponents' weaknesses. The careful application of defensive tactics can significantly impact individual matches and shape long-term team strategies, underscoring football's reliance on both physical and intellectual prowess.

Rapid technological advances in football have ushered in an era of extensive data science studies and analysis. Companies like Hudl StatsBomb¹, STATS Perform², Tracab³, and WyScout⁴ have been at the forefront of data collection, using sophisticated techniques to gather comprehensive match details. This data accessibility has fuelled the adoption of artificial intelligence (AI) methods, including machine learning (ML) and deep learning (DL) approaches, in football analytics.

In recent years, significant advances driven by AI have enabled the evaluation of player quality and on-field activity in football [4, 5]. These developments are crucial for objectively evaluating players, influencing club performance, and ensuring the financial sustainability of clubs. Metrics such as On-Ball Value (OBV) [6], Valuing Actions by Estimating Probabilities (VAEP), and eXpected Threat (xT) have emerged as key tools for analyzing various aspects of football performance [7, 8]. Despite their utility, these metrics fail to capture and evaluate defensive actions effectively. To address this gap, our research integrates established evaluation methods with deep learning techniques to improve the evaluation of defensive contributions. This study highlights ongoing efforts in football analytics to provide a more comprehensive assessment of player performance, with a particular focus on defensive skills.

The integration of advanced analytics has expanded football analysis beyond match performance evaluation to encompass comprehensive player profiling. Organizations like the CIES Football Observatory⁵ have pioneered methodologies emphasizing detailed scouting and team development metrics. A prominent approach involves assessing key technical indicators across a substantial sample of players from various European leagues [9]. Distinct technical profiles are identified, facilitating the categorization of players based on their on-field actions. This categorization helps scouts and analysts align players' abilities with specific tactical roles within a team framework. Another significant advancement is the concept of 'experience capital,' which combines playing time, team quality, and competition level to evaluate a player's development and readiness for higher-tier football [10]. This approach efficiently identifies young talents whose potential often exceeds that of more experienced but static players. By assessing competitive experience, clubs can strategically allocate resources to individuals with promising growth trajectories. Role-based profiling adds further precision to scouting by delineating the technical and tactical requirements of specific positions [11]. This method enables teams to refine their recruitment strategies and optimize squad composition by understanding the nuanced demands of each role. Lastly, methodologies for comparing technical performance, such as those presented in [12], enhance the objectivity of player evaluations by incorporating both offensive and defensive actions [13]. This

¹ <https://statsbomb.com/>

² <https://www.statsperform.com/>

³ <https://tracab.com/>

⁴ https://www.hudl.com/en_gb/products/wyscout

⁵ <https://football-observatory.com/>

includes analyzing defensive skills such as interceptions, tackles, and clearances, providing a more comprehensive view of a player's contributions.

Despite the rapid progress in football analytics, significant limitations persist in evaluating defensive actions. First, the assessment of defensive performance faces challenges due to incomplete or inadequately captured data compared to offensive actions, such as goal attempts, assists, or passes directly leading to scoring opportunities [14]. Moreover, most existing machine learning-based frameworks in football analytics predominantly focus on offensive actions or general performance metrics, leaving defensive-specific activities, such as interceptions, tackles, and clearances, relatively under-explored or insufficiently modeled [15]. This study addresses the research gap by introducing a novel approach that integrates widely recognized football metrics, such as VAEP, OBV, and xT, with advanced modeling techniques, including XGBoost and deep neural networks. Compared to state-of-the-art defensive valuation methods, our framework uniquely emphasizes the sequential nature of defensive actions, enhancing the ability to accurately capture the implications of individual defensive interventions and errors on subsequent match scenarios. Another critical constraint in football analytics arises from the substantial variability across leagues, including differences in playing styles, tactical complexity, physicality, and overall competitiveness [16, 17]. Models and analytical insights derived from datasets based on a single league may not readily generalize to other leagues or competition contexts. Consequently, conducting analyses within specific leagues or clusters of similar leagues is essential before cautiously generalizing results or transferring analytical methods to different competitive environments [18, 19]. Our research uses data from the Polish Ekstraklasa league as a detailed case study. While our results demonstrate the effectiveness of the proposed methodology within this particular league context, further comparative studies involving data from other European leagues would be beneficial to validate the generalizability and applicability of our framework across diverse football settings.

Related work

Machine learning has attracted significant attention in sports analytics, specifically revolutionizing training, tactical planning, and player health monitoring [20, 21]. This transformation is particularly evident in football, where the need for data-driven decision-making has become increasingly evident. Based on robust data science research and an influx of detailed match data from providers such as Hudl StatsBomb, and Opta, the last decade has seen technological growth in football analytics [22, 23]. This growth has accelerated the integration of various artificial intelligence methods, such as machine learning, reinforcement learning, neural networks, and deep learning, in both academic and professional settings.

In football analytics, the primary focus has been on player performance metrics, specifically expected goal (xG) and expected threat (xT) models. The xG metric aims to quantify shot quality, and numerous studies and data companies continue to refine methods for calculating shot quality [24]. In contrast, the xT model, introduced by Singh [8], employs a Markov model to analyze ball possession dynamics, offering insight into how individual actions on the pitch contribute to creating goal-scoring opportunities. However, these models, xG and xT, often focus on personal actions and overlook the interconnected nature of the events that lead to those actions. Some researchers advocate for

a more holistic approach, analyzing play sequences to understand the game's dynamics better [25–27]. This approach highlights the intricacies of football, where actions that directly lead to goals or assists are only a fraction of the total. It emphasizes the importance of player creativity and strategic decision-making. Recent methodologies have broadened this scope of analysis to include aspects such as player creativity [28], team performance evaluation, and pattern recognition in games [29].

Pattern recognition has become a key element in understanding the game's dynamics, with techniques such as clustering [30] and neural networks [31] used to identify and analyze typical sequences of actions that lead to shots or goals. Decroos et al. [15] used 'sequence pattern mining' to discover and analyze patterns in football action sequences, enriching our understanding of game tactics. Building on this holistic approach, Pappalardo et al. [16] introduced PlayeRank. This data-driven framework offers a multifaceted, role-specific evaluation of player performance, highlighting both offensive and defensive nuances of the game.

Defensive performance, historically difficult to quantify, has gained visibility through advanced machine learning and data analytics, recognizing the key role of defense in football and promoting a more balanced view of player contribution and team dynamics. This shift towards a comprehensive analytical perspective underscores the importance of studying defensive behaviors, which, although less understood than offensive actions, are crucial for accurate game strategy analysis. Integrating these methods into football analysis underscores the sport's evolving complexity and the invaluable role of machine learning in uncovering its nuanced dynamics.

Dataset and methods

Dataset

The dataset utilized for this research was primarily sourced from the Hudl StatsBomb⁶ service. It includes event data for the 2021/2022 and 2022/2023 Polish Ekstraklasa league seasons (a total of 612 games). The Hudl StatsBomb data provides a rich array of details about every match, including team and player information, possession chains, individual player actions, and event locations, which gives a holistic view of the game's dynamics.

Event data, a vital characteristic of this dataset, refers to the logged actions during a match. This includes every pass, shot, tackle, and dribble, along with the time and location of each event on the pitch. The information about each action's type, its result, and the players involved is crucial in deriving meaningful metrics, such as xT, VAEP, or OBV (Figure 4).

Additionally, the dataset is in the Soccer Player Action Description Language (SPADL) format. SPADL is a standard and universal way of representing football actions, which simplifies the process of data analysis and model building [15]. It allows for easier comparisons across different datasets and facilitates the development of data-driven insights into player and team performance. By translating raw data into a format that is easy to understand and analyze, SPADL enhances the effectiveness of statistical analysis in football.

To compute the xT values for events, we defined a set of moving actions and subsequently refined our dataset to include these specified actions. Recognizing the

⁶<https://statsbomb.com/>

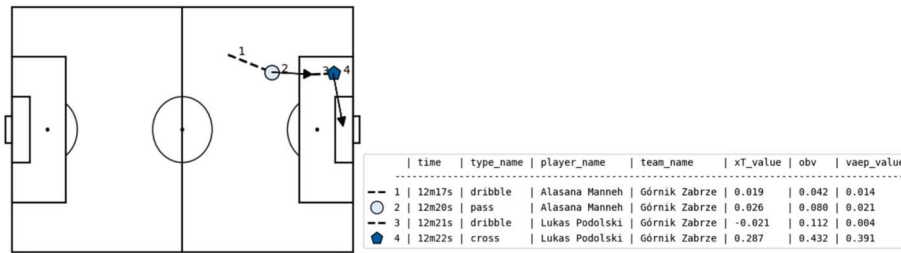


Fig. 1 Sequence of events showing evaluation metrics of each action

Table 1 Dataset information for different seasons. In red, we have pure defensive actions. In grey, we have passes – only non-defensive actions used in our approach

Events:	2021/2022	2022/2023
	579,229	572,942
Moving actions:	442,965	447,151
Clearances (C)	13,569	13,437
Crosses	2,000	2,073
Dribbles	226,942	227,315
Passes (P)	188,557	192,736
Shots	691	638
Throw ins	11,206	10,952
Interceptions (I)	6,177	4,278
Tackles (T)	2,696	2,471

significance of event sequences in football analytics, we adopted a sequence definition in alignment with the methodology outlined in the *DaxT* article [32]. A sequence is characterized by two consecutive successful events followed by an unsuccessful third event, which is then succeeded by a defensive action executed by the opposing team. Given a series of events $E_1, E_2, \dots, E_n$, a sequence S can be defined as:

$$S(i) = \begin{cases} 1 & \text{; if } E_i \text{ is successful,} \\ 1 & \text{if } E_{i+1} \text{ is successful,} \\ 1 & \text{; if } E_{i+2} \text{ is unsuccessful,} \\ 1 & \text{if } E_{i+3} \text{ is a defensive action} \\ & \text{by the opposing team,} \\ 0 & \text{otherwise,} \end{cases}$$

where $S(i)$ represents the binary outcome of the sequence starting at event E_i .

To improve the interpretability of this definition, we provide a graphical representation of a typical sequence in Figure 1. The visualization illustrates the spatial progression of consecutive actions on the pitch, accompanied by their corresponding metric values. In the accompanying table, each action is linked with its xT, OBV, and VAEP values, which quantify its offensive potential and contextual importance. For example, the cross (action 4) yields the highest values across all three metrics, indicating that the danger increases as the sequence progresses. This joint view clarifies how sequences are evaluated and how defensive actions can prevent high-value outcomes.

After defining a sequence, we determined which events should be categorized as defensive actions. Initially, this included tackles and interceptions (Table 1). We also considered fouls, given their potential tactical importance in preventing goals. However, our data contained only around 200 instances (less than 0.1% of events), making them statistically negligible at the aggregate level. We examined different models and seasons,

and the inclusion of fouls did not materially change the main outcomes. Although fouls may play a role in specific match situations, their rarity prevented them from having a measurable impact on the overall results. Therefore, we excluded fouls from the final metric formulas.

The frequency of moving actions in the dataset is presented in Figure 2.

In our dataset for the 2022/23 season, we have 348 players. However, after removing goalkeepers (24) and players who played less than 900 minutes (53), we obtained 271 players. The threshold of 900 minutes is arbitrary, but in our opinion, the player who has played 900 minutes provides us with enough information to evaluate them. The following analyses of players are limited to these 271 players. For model development, the dataset was randomly divided into training and test subsets with an 80/20 split, ensuring that reported results are based on independent test data and not on the training set. A simple summary of the used data to compare players is presented in Table 2.

Modeling input

Our framework employs three separate predictive models, one for each of the offensive metrics: xT , $VAEP$, and OBV . In other words, we train an independent model to estimate the counterfactual value of each metric if the defensive action had not occurred. This design allows us to preserve the specific interpretation of each metric while maintaining methodological consistency across models.

Formally, the model predicts the value of the next event in a possession sequence, given that a defensive intervention (interception, tackle, or clearance) was not performed. The input features are derived from the two preceding actions in the sequence. For each action i , we use its pitch coordinates (x_i, y_i) and its corresponding offensive metric value (either xT_i , $VAEP_i$, or OBV_i , depending on the model). Thus, the input vector for the model consists of:

$$(x_1, y_1, \text{metric}_1, x_2, y_2, \text{metric}_2),$$

where $(x_1, y_1, \text{metric}_1)$ represents the location and metric value of the first preceding action, and $(x_2, y_2, \text{metric}_2)$ corresponds to the second one.

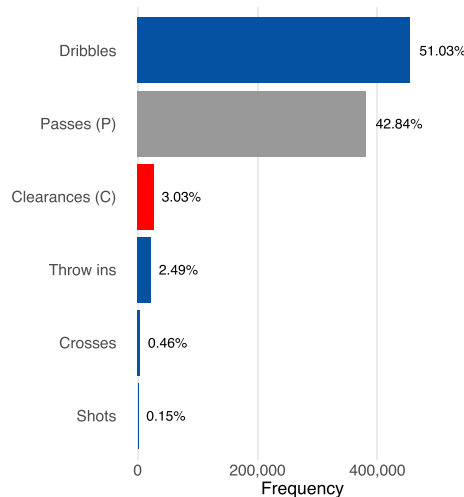


Fig. 2 Frequency of moving actions in the dataset (both seasons). In red, we have clearances – only defensive actions in moving actions. In grey, we have passes – only non-defensive actions used in our approach

Table 2 Means of used features in player evaluation (the highest values are bolded)

Position (count)	Age (years)	Weight (years)	Height(m)	Minutesplayed	SofaS- corerat- ing	Valuation(mln €)	Clear- ances	In- ter- cep- tions	Tack- les
Att. Mid- fielder (8)	27.0	69.1	1.78	1951	6.93	0.93	11.1	11.7	25.3
Centre Back (68)	28.1	79.2	1.87	2078	6.90	0.59	77.6	33.2	28.6
Cen. Def. Mid- fielder (67)	28.5	77.5	1.79	1893	6.89	0.68	14.6	4.6	14.7
For- ward (30)	28.5	77.5	1.85	1893	6.89	0.89	14.6	4.6	14.7
Full Back (62)	28.2	72.5	1.79	1677	6.80	0.55	33.6	21.7	29.1
Winger (41)	26.8	68.9	1.75	1735	6.83	0.84	8.9	10.6	24.5
Total (271)	27.8	74.0	1.81	1859	6.87	0.68	35.3	21.3	28.1

Table 3 Example input parameters for OBV model

x_1	y_1	OBV_1	x_2	y_2	OBV_2
51.66	34.04	0.0221	62.69	38.12	0.301

Before training, all input features were rescaled to the [0, 1] range using min–max normalization. While XGBoost does not require feature normalization, this step was necessary for the neural network models to ensure stable convergence and consistent feature scaling. For comparability, we used the same normalized inputs across all modeling approaches.

This sequential structure ensures that the model captures both the spatial context and the dynamic development of possession leading up to a defensive action. Table 3 provides a detailed example of the feature set for one of the metrics.

Defensive actions

The analysis of defensive actions in players considers various moments and phases of the game. Defensive activities, such as off-the-ball movements in low, middle, and high zones, are thoroughly examined. A critical aspect is the proper positioning of a player without the ball, which can significantly benefit offensive actions through secure backup. Transitional phases and set pieces are also analyzed, with frequent events including one-on-one duels, tackles, clearances, interceptions, and aerial duels. These aerial encounters are crucial in defense, requiring the player to possess physical fitness, make quick situational assessments, and respond effectively. Proper defensive positioning and the ability to compete aerially are vital in preventing dangerous and valuable offensive plays by the opponent. In our experiments, we used three pure defensive actions (Figure 3):



Fig. 3 Defensive actions. From left: clearance, interception, and tackle

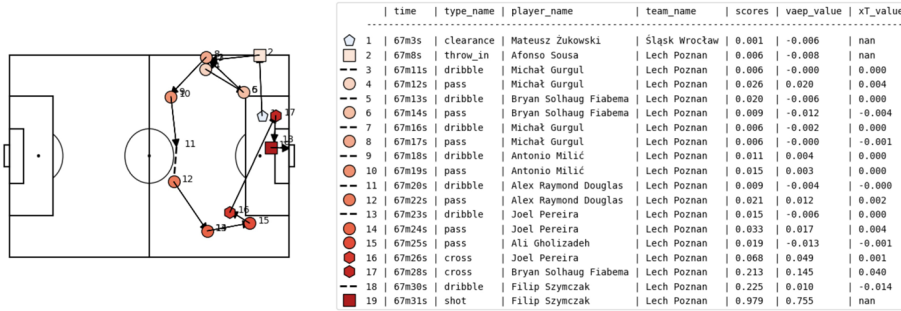


Fig. 4 Passage of play showing evaluation metrics of each action

- 1 Clearance: Kicking or heading the ball away to eliminate the danger.
- 2 Interception: Winning possession by cutting out a pass without making contact with the opponent.
- 3 Tackle: Physically winning the ball from an opponent.

Football metrics

As sports become more complex, individual metrics alone may not fully capture a player's performance. Data analysts are now combining metrics such as xT, VAEP, and OBV into a composite score to provide a holistic view of a player's contribution, including their goalscoring threat and value in possession. This method overcomes the potential bias of focusing on a single metric. It better reflects midfielders' different roles, who may excel in areas such as VAEP and OBV despite having lower xT scores. Aggregate scoring provides a more accurate measure of a player's overall impact on their team.

Expected Threat (xT)

The xT metric measures a player's likelihood of scoring a goal or creating a scoring opportunity for their team. It considers various factors, such as the player's position on the field, the type of action they are performing, and the ball's location. The xT metric enables the comparison of players across different positions and roles.

$$xT_{x,y} = \mathbb{P}_{x,y}(S) \cdot \mathbb{P}_{x,y}(G) + \mathbb{P}_{x,y}(M) \cdot \sum_{z=1}^{16} \sum_{w=1}^{12} \mathbf{T}_{(x,y) \rightarrow (z,w)} \cdot xT_{z,w},$$

where $\mathbb{P}_{x,y}(S)$ represents the shooting probability from position (x, y) , $\mathbb{P}_{x,y}(G)$ is the probability that this shot is scored, $\mathbb{P}_{x,y}(M)$ indicates the probability of ball movement actions instead of shooting ($\mathbb{P}_{x,y}(S) + \mathbb{P}_{x,y}(M) = 1$), and $\mathbf{T}_{(x,y) \rightarrow (z,w)}$ is the transition matrix that denotes the probability of the ball moving from position (x, y) to position (z, w) , with $xT_{z,w}$ being the expected threat value of the target position. The xT metric

is beneficial for evaluating the performance of attackers. The metric strongly correlated with the number of goals and chances created by these players.

Valuing Actions by Estimating Probabilities (VAEP)

VAEP is designed to evaluate the value a player adds to their team by performing various actions on the field. The VAEP metric considers the probability of a specific action leading to a goal and assigns a value to each action based on that probability.

$$V(S_i) = \mathbb{P}_{\text{score}}^k(S_i, t) - \mathbb{P}_{\text{concede}}^k(S_i, t),$$

where $\mathbb{P}_{\text{score}}^k(S_i, t)$ and $\mathbb{P}_{\text{concede}}^k(S_i, t)$ are the probabilities that team t which possess the ball in state S_i will score or concede in the next k actions. The primary distinction between the xT and VAEP metrics lies in their scope of evaluation. While xT focuses solely on quantifying the value of events involving ball movement, VAEP offers a more comprehensive analysis by considering the game's current state, such as the scoreline and the context of the three preceding actions. VAEP also factors in additional elements, such as distance and angle to the goal, the pace of the game, and remaining time, providing a more nuanced assessment of each play [26].

On-Ball Value (OBV)

OBV, a metric developed by StatsBomb in 2021, measures the effect of a player's actions on the chances of scoring or preventing goals in football. Unlike traditional metrics, OBV evaluates actions based on their contribution to overall team success, taking into account the context and outcome of each action. It uses a complex calculation method that incorporates eXpected Goal (xG) but uniquely does not attribute value to the recipients of passes. Instead, it emphasizes the value contributed by the player initiating the action. One limitation of OBV is that the details of its calculation are proprietary, so it can only be applied to data from Hudl StatsBomb [6].

Machine learning methods

Two prominent machine learning techniques, XGBoost and deep learning, were employed to evaluate players based on their defensive actions.

XGBoost

In machine learning, boosting is a highly renowned concept. Boosting functions by iteratively implementing an algorithm on differently weighted segments of the training data and then deriving a final decision based on the weights from this series of methods. Gradient boosting algorithms primarily concentrate on the discrepancies between the actual predictions and the accurate data. XGBoost [33] is an ensemble machine-learning algorithm rooted in decision trees, operating within a gradient-boosting framework. It is vital in football analytics, e.g., [20, 34, 35].

Deep learning

Deep learning is a subfield of machine learning that utilizes multi-layered neural networks to model complex patterns in data. It is particularly valuable in sports data analysis, as it can efficiently process and interpret large volumes of data, including image and time series data commonly found in sports. This enables the precise analysis of player

performance, identification of gameplay patterns, and prediction of outcomes, which can significantly contribute to optimizing training and game strategies (see, e.g., [36–38]).

Aggregation (ensemble) methods

Aggregation functions are mathematical methods used to combine multiple inputs into a single output. In ensemble learning, these functions play a crucial role in integrating predictions from various models to produce a unified, final prediction. Ensemble learning is a machine learning paradigm that combines multiple models, often of varying types, to solve a specific problem. This approach is based on the idea that a group of weak learners can collectively form a strong learner, thereby improving the predictive performance beyond what any individual model could achieve alone.

The essence of ensemble methods lies in aggregating the predictions from individual models. There are several aggregation functions, each with a distinct method for combining inputs. The choice of aggregation function can significantly influence the performance of the ensemble system. In this article, we use the following aggregation methods:

- 1 **Simple Averaging** – the predictions of all models are averaged to obtain the final prediction. This method is effective in reducing variance and is less sensitive to outliers.
- 2 **Weighted Averaging** – similar to averaging, weights are assigned to the predictions of each model, allowing the ensemble to prioritize the outputs of better-performing models.
- 3 **Ordered Weighted Averaging (OWA)** – is a sophisticated approach that combines the predictions of various models based on a set of weights that reflect the importance or reliability of each prediction in a specific order. Unlike simple or weighted averaging, where weights are directly associated with models, in OWA, the weights are assigned to the positions of the models' outputs after they have been ordered according to a specific criterion, typically their value.

An aggregation function can be formally defined as follows. Let $X = \{x_1, x_2, \dots, x_n\}$ be a set of input values. An aggregation function f is a function that maps the set X to a single output value y , such that: $f : \{x_1, x_2, \dots, x_n\} \rightarrow y$. The specific form of f depends on the type of aggregation being performed, such as:

- Simple Average (mean). The function

$$f_{\text{avg}}(X) = \frac{1}{n} \sum_{i=1}^n x_i$$

calculates the mean of the input values.

- Weighted Averaging. The function

$$f_{\text{wavg}}(X) = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i},$$

where w_i represents the weight assigned to each input value x_i , emphasizing the contribution of different inputs based on their importance or reliability.

- Ordered Weighted Averaging (OWA) [39, 40]. The function $f_{\text{owa}}(X)$ involves assigning weights to the input values based on their ranked positions after sorting. Let $X' = \{x'_1, x'_2, \dots, x'_n\}$ be the ordered set of X such that $x'_1 \leq x'_2 \leq \dots \leq x'_n$. Then,

$$f_{\text{owa}}(X) = \sum_{i=1}^n w_i x'_i,$$

where w_i are the weights associated with the ranks of the inputs, not the inputs themselves, ensuring that $\sum_{i=1}^n w_i = 1$.

Player ranking formulas

To synthesize the individual metrics gathered for Interceptions and Tackles into a comprehensive assessment of player performance, we employed three distinct aggregation methods to derive a final ranking score for each player.

The first method draws inspiration from the established formula presented in the *DaxT* [32] article. This approach delineates metrics into defensive and offensive categories, then strategically weights them to reflect their respective contributions to player performance. The weighting scheme is carefully calibrated to ensure that each metric is proportionately represented in the final score, thus providing a balanced evaluation of a player's on-field impact. The formula applied is as follows:

$$\text{DPV} = \frac{\sum_{k \in \{\text{xT}, \text{VAEP}, \text{OBV}\}} (I_k + T_k + C_k)}{9} + \frac{\sum_{k \in \{\text{xT}, \text{VAEP}, \text{OBV}\}} P_k}{3}.$$

In formula DPV denotes the final defender score. Here, I_{xT} represents the value of xT that the interception event prevented, while T_{xT} corresponds to the xT value that the tackle event prevented. The variables I_{VAEP} and T_{VAEP} denote the VAEP values stopped by interception and tackle events, respectively. Similarly, I_{OBV} and T_{OBV} represent the OBV values halted by the respective defensive actions. C_{xT} signifies the cumulative xT value from clearance events, and P_{xT} signifies the Pass xT, similarly for VAEP and OBV. The division by nine computes the mean of the defensive contributions, while the final division by two averages all the metrics involved. This formula meticulously combines defensive and offensive metrics to quantify a player's overall impact on the game.

The second implemented aggregation method is Principal Component Analysis (PCA), a statistical procedure that utilizes orthogonal transformation to convert a set of potentially correlated variables into a set of linearly uncorrelated variables known as principal components. This transformation is conducted so that the first principal component accounts for the largest possible variance in the data, with each succeeding component having the highest variance possible under the constraint of being orthogonal to the preceding components. Through PCA, we can distill the essence of the data into a set of new uncorrelated variables, each representing a dimension of the player's performance. This method is particularly advantageous in reducing the dimensionality of the data while preserving as much variability as possible, making it an invaluable tool for nuanced player evaluation and comparison. The method is applicable when there are many linear correlations in the dataset. Figure 5 presents the Pearson correlation coefficients. We observe many significant correlations, which justify the method.

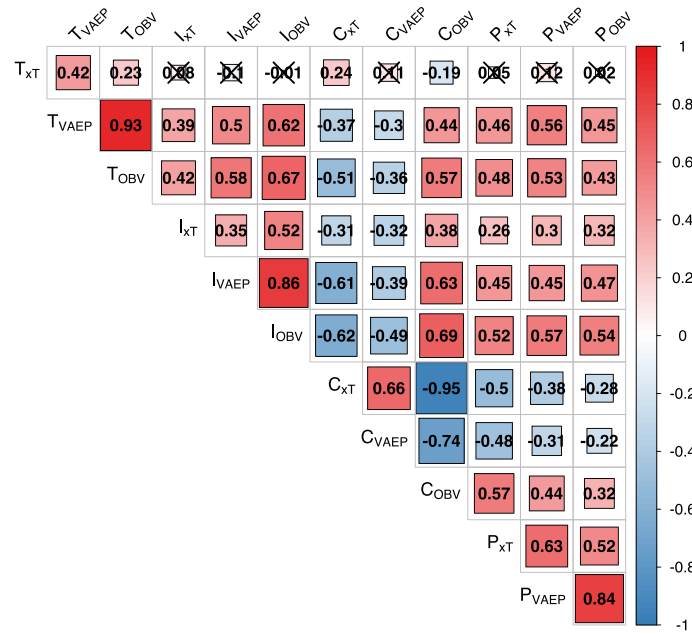


Fig. 5 Pearson correlation coefficients, insignificant (at 5%) correlations are crossed

As a third aggregation approach, we applied the Ordered Weighted Averaging (OWA) operator. This approach was based on the assumption that our focus lies on a footballer's highest metric values, to which we assign the most significant weights while disregarding the metrics where the footballer's performance is lacking. The weight vector applied in this method was defined as

$$w = [0.3, 0.2, 0.2, 0.1, 0.1, 0.05, 0.05, 0, 0, 0, 0, 0].$$

We tested several alternative OWA weight vectors and observed no substantial differences in the outcomes. The chosen vector was adopted because it prioritizes higher-ranked criteria while still accounting for the lower-ranked ones, and it provides a clear and interpretable specification.

A min-max normalization (rescaling)

$$x_{\text{new}} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

was applied to normalize all input features to the interval $[0, 1]$ before performing PCA and OWA, ensuring consistency and comparability across metrics with different ranges. This preprocessing step eliminates the bias caused by varying scales, ensuring that each metric contributes proportionally to the analysis.

Baselines

To fairly compare our solutions, we selected three baseline metrics. Baseline metrics are crucial for machine learning models because they serve as a reference point for comparing model performance, enable the quantitative assessment of improvement over a simple model, and help establish realistic performance expectations. Moreover, a simple baseline model can effectively challenge more complex models, demonstrating their value. These metrics help prevent overengineering and focus on simpler, more effective

solutions. They are cost-effective before investing in more complex and expensive models, aid in better understanding the problem at a fundamental level, and set clear goals for more complicated models.

Expert system

A crucial component of player evaluation is the expert analysis carried out by a team of experienced scouts/observers. Their role is to assess players' skills and potential, particularly aspects that are not readily apparent from existing statistical data. The study involved 24 people with experience evaluating players from professional football leagues. The study did not involve personal or sensitive data. All expert participants provided informed consent, and the procedures complied with institutional ethical standards. According to the regulations of our institution, IRB approval was not required for this type of research. Within the PKO BP Ekstraklasa league, 44 players were selected and rated on a scale of 1 to 10 based on their performance during the 2022/2023 season. However, due to injury, one of them, Luís Mata from Pogoń Szczecin, was removed from further analysis because of the cut-off value of 900 played minutes.

Player valuation

One of the criteria used to rate players is their market value, which is influenced by several factors. These include the cost of recent transfers, league and cup game appearances, national team event participation, and individual performance data. A prominent source of such data is the Transfermarkt service⁷, which offers an extensive database of football players. This database includes not only basic statistics but also market valuations of players. The information provided by Transfermarkt is widely utilized for comparisons in academic studies and by the analytical departments of football clubs. The analysis conducted included the players' market value, based on data published by Transfermarkt as of November 16, 2022, a date chosen because it followed approximately 10 league matches played by most players. To evaluate changes in value, the first valuation published by Transfermarkt after the analyzed season's conclusion (June 9, 2023, marking the end of the 2022/23 season) was used as a reference point.

SofaScore ratings

SofaScore⁸ is an application that provides real-time scores and statistics for various sports, particularly football. This platform is a numerical system crucial for rating a player's performance in individual matches. This rating objectively derives from the player's match-specific statistics, although the calculation method remains proprietary. The approach facilitates an unbiased assessment and comparison of players.

Selection of the regression method

Neural network architecture

In addressing the problem of predicting the expected threat of football events, we implemented a deep neural network (DNN) architecture. The model was implemented using the Keras library [41]. The model's architecture was designed to learn patterns within the data. It consists of multiple layers with Leaky ReLU (Leaky Rectified Linear Unit)

⁷ www.transfermarkt.pl

⁸ www.sofascore.com

Table 4 Best hyperparameters: DNN

Hyperparameter	Value
Units	128
Dropout	0.1
Number of Layers	5
Learning Rate	0.0001
Tuner Epochs	10
Tuner Initial Epoch	0
Tuner Bracket	0
Tuner Round	0

Table 5 Best hyperparameters: XGBoost

Hyperparameter	Value
Subsample	0.6
Number of Estimators	500
Minimum Child Weight	3
Maximum Depth	6
Learning Rate	0.02
Gamma	0
Colsample by Tree	0.5

activation functions to introduce non-linearity and dropout layers to avoid overfitting. The number of units in each layer and the dropout rate were treated as hyperparameters to be optimized via a tuning process using KerasTuner [42]. The network was trained using the Root Mean Squared Error (RMSE) loss function:

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{N}},$$

where y_i is the actual Metric value for the i -th observation, $\hat{y}_i$ is the predicted Metric value for the i -th observation, N is the number of observations, and $\text{Metric} \in \{\text{xT}, \text{VAEP}, \text{OBV}\}$.

The model optimization was performed using the Adam optimizer, with the learning rate also treated as a hyperparameter. This allowed for a comprehensive search over the hyperparameter space to identify the optimal learning rate in conjunction with the architecture's other parameters. The training process was regulated by a specific number of epochs (20), with early stopping implemented to prevent overfitting. The early stopping callback monitored the validation loss and stopped the training if no improvement was observed for ten consecutive epochs, ensuring that the best model weights were restored. The best hyperparameters are in Table 4.

XGBoost architecture

To complement our neural network approach and enrich our analysis, we incorporated an additional model comparison using the XGBoost [33] algorithm. The XGBoost model was initialized to minimize the squared error, which aligns with our goal of accurately predicting the xT, VAEP, and OBV. A randomized search was employed to optimize the hyperparameters over a predefined grid. The optimal parameters were searched using a 4-fold cross-validation to ensure the model's robustness and generalizability. The best hyperparameters are in Table 5.

Table 6 Comparison of RMSE for xT, VAEP, and OBV metrics

Metric	DNN	XGBoost
OBV	0.03671	0.03620
VAEP	0.03789	0.03112
xT	0.00344	0.00343

Table 7 Metrics for the top 10 players from the 2022/23 season (sorted in decreasing order of DPV). Symbols in the last column indicate increasing (▲), decreasing (▼), or no change (◆) player valuation during the examined period

No.	Player name (age, position)	Club name (league position)	DPV metric	Expert assessment	SofaScore rating	Player valuation (mln €)
1.	Damian Dąbrowski (31, CDM)	Pogoń Szczecin (4)	0.744	7.100	7.38	1.20 ▼
2.	Joel Pereira (27, FB)	Lech Poznań (3)	0.723	6.765	7.34	2.00 ◆
3.	Bartosz Slisz (24, CDM)	Legia Warszawa (2)	0.696	8.091	7.04	1.50 ▲
4.	Virgil Eugen Ghită (25, CB)	Cracovia Kraków (7)	0.692	6.600	7.04	1.50 ▲
5.	Łukasz Łakomy (22, CDM)	Zagłębie Lubin (9)	0.683	6.692	7.15	1.00 ▲
6.	Fran Tudor (28, FB)	Raków Częstochowa (1)	0.680	7.737	6.95	2.50 ▲
7.	Robert Ivanov (29, CB)	Warta Poznań (8)	0.680	6.889	7.03	0.55 ▲
8.	Erik Janža (30, FB)	Górnik Zabrze (6)	0.665	7.333	7.32	1.00 ▼
9.	Benedikt Zech (33, CB)	Pogoń Szczecin (4)	0.661	7.308	6.96	0.40 ◆
10.	Krystian Geteringer (35, FB)	Stal Mielec (11)	0.653	6.125	7.03	0.30 ▼

Results

Machine learning models

Finally, after thoroughly comparing the performance metrics between the Neural Network and XGBoost models, it was decided that the final predictions would be made using the XGBoost framework. This decision was grounded on the empirical evidence that XGBoost outperformed the Neural Network regarding the validation loss RMSE function across multiple models, as illustrated in Table 6. Furthermore, it was observed that XGBoost exhibited a significant advantage in training and hyperparameter tuning speed. Additionally, XGBoost tends to be faster in inference than neural networks [43]. The efficiency of XGBoost, combined with its predictive performance, makes it a superior choice for our application, particularly in scenarios where model training and re-tuning are frequently required.

Importantly, all results were evaluated on independent test sets. To ensure that the models were not overfitting, we monitored the difference between training and test errors throughout the experiments. The RMSE values on the test sets remained close to those on the training data, and no systematic divergence was observed. For the neural networks, early stopping was applied when the validation loss stopped improving, while for XGBoost, cross-validation and regularization parameters were used to control model complexity. These checks confirmed that the reported performance reflects genuine generalization rather than overfitting.

DPV

Table 7 presents the results for the top 10 players in the league, as determined by the proposed DPV metric. The results are compared with the expert ratings discussed above (standard deviation in parentheses in Table 7), the SofaScore ranking, and the player valuation. The table also includes information on the player's age and position (CB – Centre Back, CDM – Central Defensive Midfielder, FB – Full Back, AM – Attacking Midfielder, WB – Winger, FW – Forward), the team he played for, and, in brackets, the position in which the team finished the season. Players with a score in the top 3 for each metric are shown in light blue (expanded statements in Appendix A). Figure 6 presents the

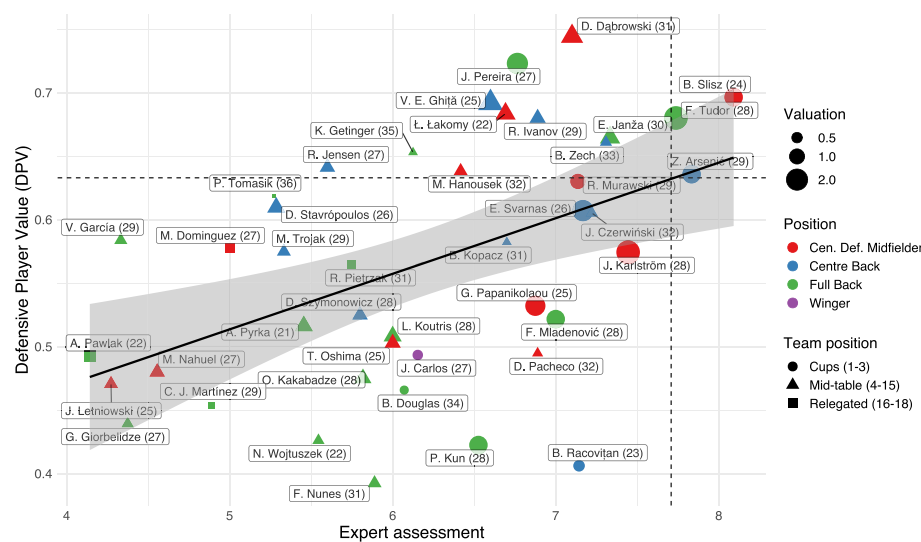


Fig. 6 Relationship between expert assessments and Defensive Player Value (DPV). Dashed lines mean 95% value of presented measures, and the solid black line is a linear regression model (the grey area is 95% confidence interval for regression)

Table 8 Metrics for the top 10 players from the 2022/23 season (sorted in decreasing order of PCA). Symbols in the last column indicate increasing (▲), decreasing (▼), or no change (◆) player valuation during the examined period. The ‘-’ means no expert rated this player in the top 44 defensive players

No.	Player name (age, position)	Club name (league position)	PCA metric	Expert assessment	SofaScore rating	Player valuation (mln €)
1.	Jakub Czerwiński (32, CB)	Piast Gliwice (5)	7.72	7.22	6.91	0.40 ▲
2.	Miłosz Trojak (29, CB)	Korona Kielce (13)	7.56	5.33	6.91	0.50 ◆
3.	Virgil Eugen Ghită (25, CB)	Cracovia Kraków (7)	7.33	6.60	7.04	1.50 ▲
4.	Zoran Arsenić (29, CB)	Raków Częstochowa (1)	7.23	7.83	7.22	1.50 ▲
5.	Benedikt Zech (33, CB)	Pogoń Szczecin (4)	7.04	7.31	6.96	0.40 ◆
6.	Robert Ivanov (29, CB)	Warta Poznań (8)	7.03	6.89	7.03	0.65 ▲
7.	Bartosz Kopacz (31, CB)	Zagłębie Lubin (9)	6.57	6.70	7.00	0.30 ◆
8.	Mateusz Matras (32, CB)	Stal Mielec (11)	6.30	–	7.06	0.50 ▼
9.	Richard Jensen (27, CB)	Górnik Zabrze (6)	6.26	5.60	7.03	0.60 ▲
10.	Dimitrios Stavrópolous (26, CB)	Warta Poznań (8)	6.25	5.29	6.93	0.80 ◆

relationship between DPV and Expert assessment rating. It shows how expert assessments correlate with DPV scores. We can observe that the Expert assessment rating tends to favor offensive and more expensive players, whereas DPV accurately identifies the best defensive players. This potential bias toward offensive players in expert ratings is mitigated by DPV, which focuses exclusively on defensive contributions. A clear trend exists where higher Expert assessments align with higher DPV scores. We calculated a linear model for this relationship and obtained a significant (p -value = 0.00103) variable Expert assessment. It means simultaneously that our model is better than the null model (intercept-only model). The value of the coefficient is equal to 0.0438. This means that we can expect a 0.0438 increase in DPV for each unit increase in Expert assessment. The fit quality of the model, measured by R^2 , is 23.3%.

PCA

The results of applying PCA are summarized in Table 8. This analysis highlights the utility of PCA in reducing complexity while retaining the key information necessary for player evaluation. The classic way to visualize the results is to plot the projected points on the plane of the first two components. Such a plot is presented in Figure 7. The PCA

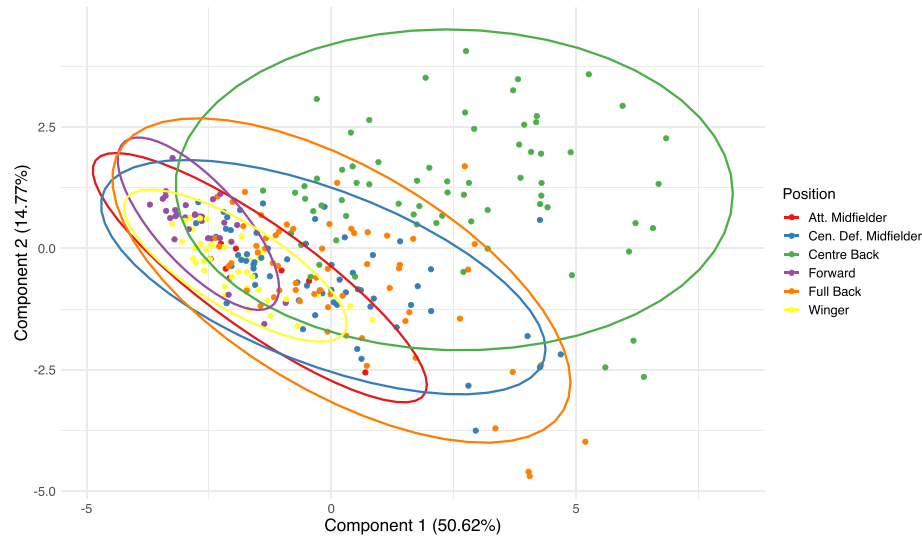


Fig. 7 The projection of players on the plane of the first two principal components with 95% normal confidence ellipses

Table 9 Metrics for the top 10 players from the 2022/23 season (sorted in decreasing order of OWA). Symbols in the last column indicate increasing (▲), decreasing (▼), or no change (◆) player valuation during the examined period. The ‘-’ means no expert rated this player in the top 44 defensive players

No.	Player name (age, position)	Club name (league position)	OWA metric	Expert assessment	SofaScore rating	Player valuation (mln €)
1.	Krystian Getinger (35, FB)	Stal Mielec (11)	0.925	6.13	7.03	0.30 ▼
2.	Joel Pereira (27, FB)	Lech Poznań (3)	0.902	6.77	7.34	2.00 ◆
3.	Víctor García (29, FB)	Śląsk Wrocław (13)	0.894	4.33	6.84	0.45 ◆
4.	Damian Dąbrowski (31, CDM)	Pogoń Szczecin (4)	0.893	7.10	7.38	1.20 ▼
5.	Zoraz Arsenić (29, CB)	Raków Częstochowa (1)	0.885	7.83	7.22	1.05 ▲
6.	Josué (33, AM)	Legia Warszawa (2)	0.881	–	7.76	1.00 ▼
7.	Fran Tudor (28, FB)	Raków Częstochowa (1)	0.881	7.74	6.95	2.50 ▲
8.	Dimitrios Stavrópolous (26, CB)	Warta Poznań (8)	0.879	5.29	6.93	0.80 ◆
9.	Jakub Czerwiński (32, CB)	Piast Gliwice (5)	0.876	7.22	6.91	0.40 ▲
10.	Miłosz Trojak (29, CB)	Korona Kielce (13)	0.863	5.33	6.91	0.50 ◆

projection illustrates the overall structure of the data in two dimensions. The observed overlap is expected due to the complexity of the feature space. It should not be interpreted as a lack of a discriminative signal, which is better captured by the predictive models. The first component explains 50.62% of the variance, and the second is 14.77%. Unfortunately, the method seems to have problems capturing positions (confidence ellipses overlap). On the other hand, the first component, which we used to measure players’ defensive quality, looks much better when differentiating players.

OWA

The Ordered Weighted Averaging operator was used to rank players based on their defensive contributions, with results summarized in Table 9. This method enables structured evaluation by assigning greater importance to high-impact actions, emphasizing players who excel in specific defensive metrics. For instance, Krystian Getinger ranked first (OWA score: 0.925), while Joel Pereira placed second (0.902), reflecting their strong performance in areas such as interceptions and clearances. Compared to PCA, OWA provides more intuitive insights by focusing on peak contributions rather than distributing weights uniformly across all variables. However, the effectiveness of OWA depends on the weight configuration, which can introduce bias if not carefully tuned.

An interesting extension of this approach involves adjusting the weight vector to control the relative influence of offensive and defensive metrics. This would enable an ablation-style analysis, revealing how variations in weighting impact player rankings and highlighting the specific importance of defensive actions in overall evaluation. Future work could explore adaptive or context-sensitive weighting schemes based on game scenarios or tactical roles.

Impact on transfers

Bartosz Slisz (Legia Warszawa), detailed in Table 7 with a DPV of 0.696 (third place), aligns with high expert and SofaScore evaluations, which may provide additional context to his transfer fee of €1.50 million and subsequent move to Atlanta United FC (Major League Soccer) for \$3.5 million. While no direct statistical causality can be established, these indicators suggest that DPV could be consistent with market trends.

Moreover, Richard Jensen, with a PCA of 6.26 (ninth place) and divergent expert and SofaScore assessments, was transferred from Górnik Zabrze to Aberdeen FC (Scottish Premiership) for approximately €550,000. This example highlights that DPV rankings do not always reflect market valuations.

Similarly, Łukasz Łakomy (Zagłębie Lubin), with a DPV of 0.683 (fifth place), received consistent expert and SofaScore ratings, and his transfer to Young Boys Bern (Swiss Super League) for about €2 million is in line with his market valuation of €1.50 million. These cases illustrate that while DPV can offer useful context, transfer fees are influenced by multiple factors beyond the presented metrics.

It is worth noting that the top ranks include many older players who, despite their high skills, face challenges with transfers and increasing their market value. For example, according to DPV, the leader is Damian Dąbrowski (Pogoń Szczecin), who is 31 years old. Based on PCA, the top player is Jakub Czerwiński (Piaś Gliwice), aged 32. Finally, according to OWA, the best defender is the 35-year-old Krystian Getinger (Stal Mielec).

Feature importance

During the creation of PCA, we obtain so-called loadings. They are the weights of the original variables used to create components. We can use their absolute values to determine the importance of original features in building components. The first component is the most important (we use it). Hence, we can use the loadings of these components as a variable importance. In Figure 8, we presented the importance of the features calculated in the way described (scaled to sum to 1). The plot shows that the remaining variables have similar importance except T_{xT} . When we add importance for each metric, we obtain the total feature importance for OBV: 0.390, VAEP: 0.357, and xT: 0.254. We can conclude that all three metrics are essential to calculating the defensive valuation of a footballer. Similarly, we can calculate the importance of Clearances: 0.276, Interceptions: 0.271, Passes: 0.258, and Tackles: 0.195. Hence, the most important are clearances, and the least important are tackles.

On the other side, the OWA helps us determine the importance of the features. We have weights for each feature for each player, and we can take the mean over these metrics to obtain the importance. In Figure 9, we presented the importance of the features calculated in the way described. When we add importance for each metric, we obtain the total feature importance for OBV: 0.517, VAEP: 0.422, and XT: 0.061. We see that OBV

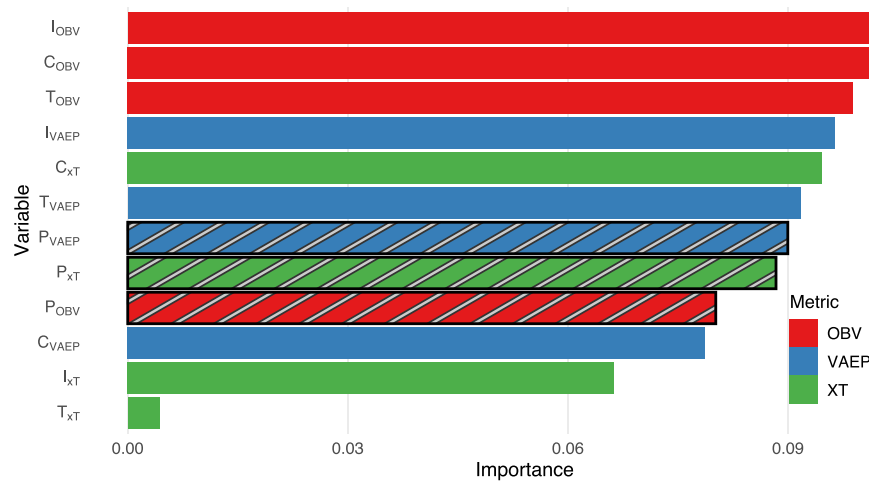


Fig. 8 Feature importance based on PCA. The passes, the only non-pure defensive stat, are outlined in black and striped

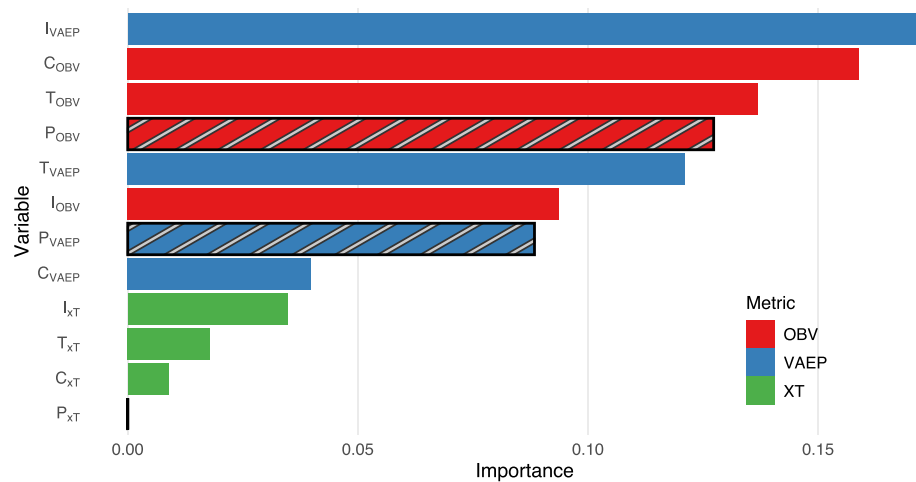


Fig. 9 Feature importance based on OWA. The passes, the only non-pure defensive stat, are outlined in black and striped

and VAEP are much more critical than xT. Similarly, we can calculate the importance of Interceptions (0.301), Tackles (0.276), Clearances (0.208), and Passes (0.216). Passes are the least important, and clearances are the most important.

The results obtained from OWA seem more reasonable. Results from PCA are not conclusive; all variables are of similar importance. OWA provides a clear conclusion: VAEP and OBV are more effective in describing players' defensive capabilities. Additionally, pure defensive statistics should be used, not passes. It is also worth mentioning that we cannot determine the importance of the features from DPV. In this approach, we only average metrics with equal weights.

Concordance of rankings

We used two widely recognized measures of ranking agreement to assess the compatibility of the proposed methods with the selected baselines.

- Spearman's rank correlation coefficient evaluates the direction and strength of the relationship between two ranked variables. Like Pearson's correlation, Spearman's coefficient ranges from -1 to 1 . The absolute value of this coefficient is considered to determine the strength of the association. Interpretations of the coefficient may vary across disciplines. Still, generally, values below 0.1 indicate no correlation, between 0.1 and 0.3 suggest a weak correlation, 0.3 to 0.5 reflect a moderate correlation, 0.5 to 0.7 indicate a strong correlation, and values above 0.7 represent a powerful correlation.
- Cohen's kappa coefficient measures the level of agreement between two raters assigning cases to mutually exclusive categories. While it identifies whether raters assess the same characteristic, it does not account for the magnitude of disagreement. We employed the weighted version of Cohen's kappa to address this limitation. Values below 0 indicate no agreement, between 0 and 0.2 reflect slight agreement, 0.2 to 0.4 signify fair agreement, 0.4 to 0.6 represent moderate agreement, 0.6 to 0.8 indicate substantial agreement, and 0.8 to 1 denote almost perfect agreement.

The results are presented in Table 10. The findings indicate that compatibility is moderate at best for SofaScore and relatively weak for valuation. The relatively low correlation with market valuations is expected, as transfer values are shaped not only by on-field performance but also by factors such as age, contract length, and reputation. This relationship is confirmed by the correlation coefficient for the valuation-age pair, which is -0.3037^* . This indicates that player valuations tend to decrease on average as age increases. Among the valuation methods, OWA exhibited the highest level of compatibility. It is worth mentioning that these metrics should therefore be viewed as performance-oriented indicators, with only indicative rather than direct financial applicability.

To assess the consistency of the rankings obtained from the three methods (PCA, DPV, OWA), we performed Kendall's coefficient of concordance (W) and the Friedman's rank sum test. The concordance analysis showed a very strong agreement between the rankings ($W = 0.954$, $p < 0.001$). Furthermore, the Friedman's test indicated no statistically significant differences between the methods ($\chi^2(2) = 1.4$, $p = 0.5$). These findings demonstrate that all three methods yield highly similar rankings of the considered alternatives.

Application of DPV for other leagues

To further test the robustness and generalizability of our approach, we extended the analysis beyond the Polish Ekstraklasa to include two additional competitions: the Portuguese Primeira Liga and the Spanish Segunda División. These leagues were selected because they differ notably in playing styles and competitive balance, allowing us to examine whether DPV remains consistent across diverse football contexts.

Table 10 Results of concordance measures (the highest values are bolded)

Method	Spearman		Cohen	
	SofaScore	Valuation	SofaScore	Valuation
OWA	0.263*	0.162*	0.263*	0.163*
PCA	0.391*	-0.015	0.392*	-0.015
DPV	0.487*	0.109	0.487*	0.110

(* means statistical significance at 5%)

In addition, we observed that in higher-quality leagues, such as the Portuguese Primeira Liga, the model achieved lower RMSE values, suggesting more stable and consistent predictions. This may be explained by the greater tactical organization and higher average technical level in these competitions, which produces more structured event sequences that our models can capture more effectively. Importantly, several players with consistently high DPV scores in these leagues secured major transfers in the following season, further underlining the external validity of the metric. In contrast, the Spanish Segunda División is characterized by a greater diversity of playing styles and more physical, less structured games, which could explain the slightly lower performance observed there. These findings indicate that while DPV is transferable across contexts, it may yield even stronger results in leagues where tactical patterns are more regular and interpretable by data-driven models.

An additional validation of the DPV metric can be seen when analyzing players from Portugal's Primeira Liga and Spain's La Liga 2 who subsequently recorded significant market value increases. Several players with high DPV scores during the analyzed seasons achieved major transfers or rapid growth in their Transfermarkt valuation in the following year. This highlights the practical usefulness of the metric for identifying young or undervalued defenders whose defensive contributions may not be fully captured by traditional statistics. Table 11 presents selected examples.

These cases illustrate that DPV is not only effective in capturing current defensive quality but also in flagging players with strong development potential. The significant increases in market valuations between 2023 and 2024 reflect how the players' on-pitch contributions, already highlighted by our metric, later translated into recognition on the transfer market. This confirms that DPV can serve as a practical tool for clubs seeking to identify and invest in defenders with high growth trajectories.

For both leagues, the concordance analysis showed a very strong agreement between the rankings (Spain 2: $W = 0.875$, $p < 0.001$; Portugal: $W = 0.850$, $p < 0.001$). Furthermore, the Friedman's test indicated no statistically significant differences between the methods (Spain 2: $\chi^2(2) = 0.875$, $p = 0.5824$; Portugal: $\chi^2(2) = 4.211$, $p = 0.1218$).

Discussion and conclusion

Discussions with experts from Lech Poznań revealed that central defenders often receive the most attention in evaluations, while full-backs are usually overlooked. This discrepancy is partly because full-backs typically play more offensive roles, which traditional defensive metrics do not adequately capture. However, the proposed Defensive Player Value metric addresses this gap by incorporating advanced metrics such as OBV and

Table 11 Selected players from Portugal and Spain 2 with high DPV scores in 2022/23 and their subsequent Transfermarkt valuation changes. The last column shows their position in our DPV ranking for the respective league

No.	Player name(age, pos, club)	Value 2023(M€)	Value 2024(M€)	Change(M€)	DPV Rank (league)
1.	Alex Grimaldo (28, LB, Benfica)	25.0	45.0	+20.0	2nd (Portugal)
2.	Pepelu (25, CDM, Levante)	3.0	22.0	+19.0	3rd (Spain 2)
3.	Gonçalo Inácio (23, CB, Sporting CP)	30.0	45.0	+15.0	1 st (Portugal)
4.	Saúl Coco (24, CB, Las Palmas)	3.0	14.0	+11.0	8th (Spain 2)
5.	Mika Märmol (22, CB, Andorra)	5.0	15.0	+10.0	1 st (Spain 2)

VAEP to ensure that defenders who contribute to offensive plays or ball distribution are correctly recognized. This approach balances offensive and defensive contributions, thereby improving the accuracy of player evaluations.

In addition to the DPV metric, PCA and OWA methods were applied to further refine player evaluations. PCA effectively reduces the dimensionality of data, summarizing a player's performance into one component while preserving most of the variance. In this study, PCA highlighted critical features influencing defensive performance, such as OBV and VAEP, and demonstrated its utility in identifying patterns across multiple correlated variables. OWA provided an alternative approach, focusing on the highest-valued metrics while de-emphasizing less significant contributions. This method prioritized players' peak performances by assigning greater weight to their strongest attributes. OWA emphasized the relative importance of metrics such as OBV and VAEP over xT in evaluating defensive capabilities, providing a more nuanced assessment of defensive contributions. Furthermore, its results align more closely with expert evaluations, emphasizing its practical application in scouting and player development.

One of the most important benefits of the proposed metric is its applicability to scouting. By providing a holistic evaluation of players' defensive and offensive skills, scouts can identify versatile talents who might otherwise be undervalued. This is especially valuable for identifying players with rare skills that contribute to defensive stability and effective offensive transitions.

Nevertheless, several constraints remain that shape the interpretation of our results. We therefore make them explicit and outline concrete next steps below.

Limitations and future directions

Our current DPV focuses on on-ball defensive outcomes available in event data (clearances, interceptions, tackles). This ensures reproducibility across common feeds but omits off-ball defense such as space denial, lane blocking, cover-shadowing, delaying, and coordinated pressing. Event-only data also limits our ability to quantify spatial coverage, pressure intensity, line height, compactness, or ball-oriented shifting, which typically require tracking. Future work will extend DPV in four directions. First, we will add tracking-derived off-ball components: (a) space denial via changes in pitch control around the ball and goal, (b) pressure-induced value measured as reductions in expected progression under pressure, and (c) cover-shadow effects captured as drops in pass success to shadowed receivers. Second, we will explicitly condition DPV on tactical context by classifying out-of-possession states (e.g., high press vs. mid/low block, rest-defense during own possession) and using features such as pressing height, block depth, line compactness, and out-of-possession width. Third, to improve cross-position comparability we will report within-position z-scores and explore hierarchical models with role-specific baselines (CB, FB, DM, etc.). Fourth, we will broaden data scope beyond a single league and add robustness checks for opponent strength, game state, and minutes played (e.g., club fixed effects, per-90 normalization), alongside sensitivity analyses over weights and subcomponents. Because location-dependent metrics such as *DaxT* are sensitive to context, future iterations will also incorporate match time, scoreline, preceding events, and pitch-sector effects to refine defensive valuation. These additions preserve the transparency of DPV while addressing key blind spots: off-ball activity, tactical-system effects, and league- and role-specific context.

The methods and results presented in this study demonstrate significant advances over existing solutions. The Defensive Player Value metric illustrates how established metrics, such as VAEP, OBV, and xT, can be combined with innovative models, including deep learning, to assess defensive actions. This approach provides a dynamic assessment of player contributions by predicting outcomes interrupted by defensive plays. This metric provides a season-long or partial-season evaluation of players' effectiveness at intercepting balls and stopping opposing attacks. While these are fundamental defensive skills, they are only part of a comprehensive defensive skill set that includes positioning, tackling, aerial battles, marking, and communication. High effectiveness in intercepting and preventing attacks significantly improves defensive organization by minimizing opponent scoring opportunities and maintaining defensive line stability. Future research should integrate these additional tributes to better assess a defendant. The proposed framework offers extensive potential for further study. It allows for detailed analysis of player performance in specific areas of the pitch and the impact of situational factors, such as the match's score. It also holds promise for tracking player development, assessing transfer eligibility, and comparing expectations across leagues. These possibilities mark an exciting direction for advancing sports analytics and refining defensive evaluation in football.

Table 12 A comparison of the DPV metric scores with other popular metrics. The top 30 players, according to DPV, plus all assessed by experts from the 2022/23 season (the ‘–’ means no expert rated this player in the top 44 defensive players). Symbols in the last column indicate increasing (▲), decreasing (▼), or no change (◆) player valuation during the examined period

No.	Player name (age, position)	Club name (league position)	DPV metric	Expert assessment	SofaScore rating	Player valuation (mln €)
1.	Damian Dąbrowski (31, CDM)	Pogoń Szczecin (4)	0.744	7.100 (± 0.85)	7.38	1.20 ▼
2.	Joel Pereira (27, FB)	Lech Poznań (3)	0.723	6.765 (± 1.25)	7.34	2.00 ◆
3.	Bartosz Slisz (24, CDM)	Legia Warszawa (2)	0.696	8.091 (± 0.92)	7.04	1.50 ▲
4.	Virgil Eugen Ghită (25, CB)	Cracovia Kraków (7)	0.692	6.600 (± 1.90)	7.04	1.50 ▲
5.	Łukasz Łakomy (22, CDM)	Zagłębie Lubin (9)	0.683	6.692 (± 0.95)	7.15	1.00 ▲
6.	Fran Tudor (28, FB)	Raków Częstochowa (1)	0.680	7.737 (± 1.10)	6.95	2.50 ▲
7.	Robert Ivanov (29, CB)	Warta Poznań (8)	0.680	6.889 (± 1.83)	7.03	0.65 ▲
8.	Erik Janža (30, FB)	Górnik Zabrze (6)	0.665	7.333 (± 1.16)	7.32	1.00 ▼
9.	Benedikt Zech (33, CB)	Pogoń Szczecin (4)	0.661	7.308 (± 1.11)	6.96	0.40 ▼
10.	Krystian Getinger (35, FB)	Stal Mielec (11)	0.653	6.125 (± 1.13)	7.03	0.30 ▼
11.	Josué (33, AM)	Legia Warszawa (2)	0.645	–	7.76	1.00 ▼
12.	Richard Jensen (27, CB)	Górnik Zabrze (6)	0.641	5.600 (± 1.65)	7.03	0.60 ▲
13.	Marek Hanousek (32, CDM)	Widzew Łódź (12)	0.638	6.417 (± 1.38)	6.90	0.50 ▼
14.	Zoran Arsenić (29, CB)	Raków Częstochowa (1)	0.636	7.833 (± 0.99)	7.22	1.50 ▲
15.	Radosław Murawski (29, CDM)	Lech Poznań (3)	0.630	7.136 (± 1.17)	6.97	1.00 ◆
16.	Piotr Tomasiak (36, FB)	Wisła Płock (16)	0.619	5.273 (± 0.91)	6.89	0.20 ▼
17.	Dominik Furman (31, CDM)	Wisła Płock (16)	0.610	–	7.14	0.60 ▼
18.	Dimitrios Stavrópolous (26, CB)	Warta Poznań (8)	0.610	5.286 (± 1.25)	6.93	0.80 ◆
19.	Efstathios Svarnas (26, CB)	Raków Częstochowa (1)	0.607	7.167 (± 1.19)	7.10	2.00 ◆
20.	Jakub Czerwiński (32, CB)	Piast Gliwice (5)	0.605	7.222 (± 1.31)	6.91	0.40 ▲
21.	Jarosław Kubicki (28, CDM)	Lechia Gdańsk (17)	0.588	–	6.94	0.60 ▼
22.	Víctor García (29, FB)	Śląsk Wrocław (15)	0.584	4.333 (± 1.86)	6.84	0.45 ◆
23.	Bartosz Kopacz (31, CB)	Zagłębie Lubin (9)	0.582	6.700 (± 1.34)	7.00	0.30 ◆
24.	Maxime Dominguez (27, CDM)	Miedź Legnica (18)	0.577	5.000 (± 1.41)	6.96	0.40 ▲
25.	Patryk Stepiński (28, CB)	Widzew Łódź (12)	0.576	–	6.99	0.75 ▼
26.	Miłosz Trojak (29, CB)	Korona Kielce (13)	0.575	5.333 (± 1.44)	6.91	0.50 ◆
27.	Jesper Karlström (28, CDM)	Lech Poznań (3)	0.574	7.444 (± 0.98)	7.03	2.50 ▲
28.	Raphael Rossi Branco (33, CB)	Radomiak Radom (10)	0.570	–	7.02	0.45 ▼
29.	Mateusz Matras (32, CB)	Stal Mielec (11)	0.570	–	7.06	0.50 ▼
30.	Rafał Pietrzak (31, FB)	Lechia Gdańsk (17)	0.564	5.750 (± 1.66)	6.89	0.35 ▼
39.	Giannis Papanikolaou (25, CDM)	Raków Częstochowa (1)	0.532	6.875 (± 1.15)	6.92	1.80 ▲
49.	Dawid Szymonowicz (28, CB)	Warta Poznań (8)	0.525	5.800 (± 0.92)	6.92	0.60 ▲
51.	Filip Mladenović (28, FB)	Legia Warszawa (2)	0.522	7.000 (± 1.56)	7.24	1.50 ▼
54.	Arkadiusz Pyrką (21, FB)	Piast Gliwice (5)	0.516	5.455 (± 1.57)	6.68	0.80 ▲
58.	Leonardo Koutiris (28, FB)	Pogoń Szczecin (4)	0.508	6.000 (± 1.35)	7.10	0.80 ◆
64.	Takuto Oshima (25, CDM)	Cracovia Kraków (7)	0.503	6.000 (± 1.56)	6.81	0.65 ▲
72.	Dani Pacheco (32, CDM)	Górnik Zabrze (6)	0.495	6.889 (± 1.62)	7.21	0.35 ◆
75.	Jean Carlos (27, WB)	Raków Częstochowa (1)	0.493	6.154 (± 1.07)	6.94	0.50 ▲
77.	Aleksander Pawlak (22, FB)	Wisła Płock (16)	0.493	4.143 (± 1.07)	6.71	0.60 ▼
86.	Matías Nahuel (27, CDM)	Śląsk Wrocław (15)	0.480	4.556 (± 1.51)	6.99	0.60 ◆
92.	Otar Kakabadze (28, FB)	Cracovia Kraków (7)	0.475	5.818 (± 1.40)	6.85	0.70 ◆
100.	Juliusz Letniowski (25, CDM)	Widzew Łódź (12)	0.471	4.273 (± 2.00)	6.81	0.50 ▼
104.	Barry Douglas (34, FB)	Lech Poznań (3)	0.466	6.071 (± 1.38)	6.98	0.40 ▼
110.	Carlos Julio Martínez (29, FB)	Miedź Legnica (18)	0.454	4.889 (± 1.54)	6.80	0.25 ▼
121.	Guram Giorbelidze (27, FB)	Zagłębie Lubin (9)	0.439	4.375 (± 1.41)	6.86	0.40 ◆
132.	Norbert Wojtuszek (22, FB)	Górnik Zabrze (6)	0.426	5.545 (± 1.04)	6.86	0.40 ◆
135.	Patryk Kun (28, FB)	Raków Częstochowa (1)	0.422	6.526 (± 0.90)	6.86	1.50 ◆
162.	Bogdan Racovițan (23, CB)	Raków Częstochowa (1)	0.406	7.143 (± 0.95)	6.75	0.60 ◆
182.	Fábio Nunes (31, FB)	Widzew Łódź (12)	0.392	5.889 (± 1.45)	6.79	0.50 ▼

Table 13 A comparison of the DPV metric with OWA and PCA for the top 30 players by DPV in the 2022/23 season of the Portuguese top division (Primeira Liga). Symbols in the last column indicate increasing (▲), decreasing (▼), or no change (◆) in player valuation during the examined period

No.	Player name (age, position)	Club name (league position)	DPV metric	OWA metric	PCA metric	SofaScore rating	Player valuation (mln €)
1.	Gonçalo Inácio (21, CB)	Sporting CP (4)	0.622	0.585	5.075	7.01	18.00 ▲
2.	Alex Grimaldo (27, FB)	Benfica (1)	0.620	0.806	5.598	7.52	20.00 ▲
3.	Matheus Reis (28, CB)	Sporting CP (4)	0.606	0.275	2.142	6.86	5.00 ▲
4.	Nicolás Otamendi (35, CB)	Benfica (1)	0.577	0.623	6.319	7.27	4.50 ▼
5.	Florentino Luís (23, POS)	Benfica (1)	0.575	0.281	2.682	7.10	8.00 ▲
6.	João Mário (30)	Benfica (1)	0.535	0.202	1.230	7.35	12.00 ▲
7.	Sebastián Coates (32, CB)	Sporting CP (4)	0.525	0.210	2.339	6.92	13.00 ▼
8.	Fredrik Aursnes (27, CDM)	Benfica (1)	0.522	0.391	2.326	7.01	7.00 ▲
9.	Otávio (28, AM)	FC Porto (2)	0.509	0.362	3.053	7.72	30.00 ▲
10.	Mateus Uribe (32, CDM)	FC Porto (2)	0.501	0.543	3.902	7.13	16.00 ▼
11.	António Silva (19, CB)	Benfica (1)	0.495	0.425	3.207	7.17	0.30 ▲
12.	Enzo Fernandez (22, CDM)	Benfica (1)	0.487	0.245	2.132	7.53	6.50 ▲
13.	André Horta (26, CDM)	Sporting Braga (3)	0.480	0.272	1.331	6.99	5.00 ▲
14.	Vitor Carvalho (26, CDM)	Gil Vicente (13)	0.478	0.263	2.377	7.08	1.50 ▲
15.	Ali Musrati (27, CDM)	Sporting Braga (3)	0.461	0.326	3.370	7.23	12.00 ▲
16.	Ricardo Horta (28)	Sporting Braga (3)	0.447	0.232	1.172	7.38	20.00 ◆
17.	Manuel Ugarte (22, CDM)	Sporting CP (4)	0.440	0.502	2.749	7.05	12.00 ▲
18.	Aderlan Santos (34, CB)	Rio Ave (12)	0.418	-0.682	3.362	7.05	0.50 ▼
19.	Sebastián Pérez (30, CDM)	Boavista (9)	0.417	0.372	2.977	6.96	1.50 ▲
20.	Filipe Relvas (23, CB)	Portimonense (15)	0.414	0.399	4.662	7.11	0.60 ▲
21.	Guga (25, CDM)	Rio Ave (12)	0.413	0.299	1.477	7.10	0.75 ▲
22.	Pepê (26)	FC Porto (2)	0.411	0.318	1.649	7.05	1.00 ▲
23.	Ivan Marcano (35, CB)	FC Porto (2)	0.409	0.112	1.522	7.16	0.50 ◆
24.	João Teixeira (29, CDM)	Chaves (7)	0.408	0.437	3.574	7.02	0.48 ▲
25.	Samú (27, AM)	Vizela (11)	0.403	0.368	3.602	7.01	0.90 ▲
26.	David Simão (33, CM)	FC Arouca (5)	0.403	0.310	2.038	6.98	0.25 ◆
27.	Vitor Tormena (27, CB)	Sporting Braga (3)	0.397	0.411	3.743	6.99	3.00 ▲
28.	Pepe (40, CB)	FC Porto (2)	0.387	0.358	2.340	7.02	1.00 ◆
29.	Raphael Guzzo (28, CDM)	Vizela (11)	0.387	0.295	2.912	6.95	0.30 ▲
30.	João Basso (26, CB)	FC Arouca (5)	0.383	0.525	6.430	7.12	0.70 ▲

Table 14 A comparison of the DPV metric with OWA and PCA for the top 30 players by DPV in the 2022/23 season in Spain's second division (LaLiga 2). Symbols in the last column indicate increasing (▲), decreasing (▼), or no change (◆) in player valuation during the examined period

No.	Player name (age, position)	Club name (league position)	DPV metric	OWA metric	PCA metric	SofaScore rating	Player valuation (mln €)
1.	Mika Mámol (21, CB)	FC Andorra (7)	0.910	0.672	5.990	7.12	0.80 ▲
2.	Jonathan Viera (33, CM)	Las Palmas (2)	0.830	0.121	1.358	7.28	2.00 ▼
3.	Pepelu (24, CDM)	Levante UD (3)	0.669	0.426	4.357	7.18	3.00 ◆
4.	Marc Aguado (23, CDM)	FC Andorra (7)	0.645	0.500	3.655	7.00	0.50 ▲
5.	Matheus Pereira (25, CM)	Eibar (5)	0.600	0.389	2.617	6.99	0.70 ▲
6.	Eric Curbelo (29, CDM)	Las Palmas (2)	0.593	0.600	5.707	7.09	0.80 ▲
7.	Anaitz Arbilla (36, FB)	Eibar (5)	0.583	0.780	7.507	7.02	0.40 ◆
8.	Saúl Coco (24, CB)	Las Palmas (2)	0.560	0.462	3.213	6.96	0.50 ▲
9.	Pablo de Blasis (35, AM)	FC Cartagena (9)	0.554	0.411	3.106	7.27	0.50 ◆
10.	Enzo Loiodice (22, CM)	Las Palmas (2)	0.553	0.322	1.951	7.06	1.00 ▲
11.	Sergi Cardona (23, FB)	Las Palmas (2)	0.539	0.908	8.614	6.72	0.80 ▲
12.	Adrià Vilanova (26, CDM)	FC Andorra (7)	0.526	0.125	0.366	6.89	0.50 ◆
13.	Alberto Del Moral (22, CDM)	Villarreal B (18)	0.515	0.427	4.037	6.91	0.80 ▲
14.	Adrián De la Fuente (24, CB)	Villarreal B (18)	0.508	0.380	4.850	7.00	0.80 ◆
15.	Oriol Rey (25, CDM)	Mirandés (16)	0.495	0.329	2.721	6.83	0.60 ▲
16.	Kelechi Nwakali (24, CDM)	Ponferradina (19)	0.486	0.520	3.384	7.17	0.70 ▲
17.	Rober (28, CB)	Levante UD (3)	0.475	0.704	5.967	6.96	1.20 ◆
18.	Alberto Moleiro (19, LM)	Las Palmas (2)	0.472	0.182	-0.338	6.81	3.00 ▲
19.	Carlo Adriano (22, CDM)	Villarreal B (18)	0.469	0.449	2.602	6.90	0.6 ▲
20.	Omenuke Mfulu (29, CDM)	Las Palmas (2)	0.468	0.186	0.579	6.75	0.80 ◆
21.	Hector Hevel (27, CM)	FC Andorra (7)	0.464	0.214	0.112	6.67	0.80 ▲
22.	Álex Suárez (30, CB)	Las Palmas (2)	0.462	0.453	2.809	6.96	0.50 ▲
23.	Riki Rodríguez (25, CDM)	Albacete (6)	0.462	0.374	2.175	6.88	0.40 ▲
24.	Manu Fuster (25, AM)	Albacete (6)	0.449	0.250	1.471	7.22	0.80 ▲
25.	Iker Undabarrena (28, CDM)	Leganés (14)	0.433	0.372	3.762	6.91	0.50 ▲
26.	Vicente Iborra (35, CDM)	Levante UD (3)	0.431	0.389	2.469	6.93	2.00 ▼
27.	Alex Febas (27, CM)	Málaga (20)	0.431	0.425	3.917	6.74	1.00 ▲
28.	Flavien Boyomo (21, CB)	Albacete (6)	0.431	0.462	4.734	6.82	0.80 ▲
29.	Juan Berrocal (24, CB)	Eibar (5)	0.430	0.300	3.217	6.91	1.00 ◆
30.	Lluís López (26, CB)	Real Zaragoza (13)	0.424	0.309	2.514	6.88	0.80 ◆

Appendix A

See tables 12, 13, 14.

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Author contributions

M.Z. conducted the full implementation of the study, including data preprocessing, model development, and evaluation. K.D. implemented the OWA aggregation method. T.P. and T.G. supervised and led the research design and direction. B.G. provided football expertise, access to data and contributed expert player rankings. T.P. also supported the acquisition of player values and ranking data. All authors contributed to writing the manuscript. T.G., T.P., and K.D. participated in reviewing and refining the final version of the manuscript.

Data availability

No datasets were generated or analysed during the current study.

Declarations

Competing interests

The authors declare no competing interests.

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